



Energy Analyst Programming Exercise – Instructions

Please read all instructions carefully and make sure you understand them. Ask any questions you have about the instructions prior to your scheduled timed response window.

The goal of this exercise is to evaluate how you conduct analysis, follow detailed written directions, write or direct code, and visualize data — including your ability to leverage modern AI tools effectively and critically.

Preparation

The historical power price data is provided directly in the exercise repository (see the *historicalPriceData* folder). Create a new, empty directory where you will save all output you generate as part of this exercise (CSV files, images, etc.). Read through the full list of task descriptions in *ProgrammingExercise_Tasks.docx* to understand the full scope of the exercise before beginning.

Your Response

You will have 1 hour to respond to this exercise. Your response includes all output you generate (CSV files, images, etc.) as well as all the code, notebooks, or other materials you produce to complete the exercise. Save all such files in an organized manner within the directory you created above. After the hour is up, submit your response by sharing all materials via a web-hosted Git repository (GitLab, GitHub, Bitbucket, etc.) or as a compressed archive sent to the cQuant staff member who contacted you.

You are encouraged to be efficient and complete as much of the exercise as possible. It is **not strictly required** that every task be fully complete for you to be eligible for hire. Complete the items marked as **Tasks** first; return to **Bonus** items only if time permits.

Use of AI Tools

AI tools are explicitly permitted and encouraged. This exercise has been redesigned to reflect how modern energy analysts work. You may use any AI assistant (e.g., Claude, ChatGPT, Gemini, GitHub Copilot, etc.) to help you write, debug, or explain code, explore the data, or generate visualizations.

However, effective AI use is itself part of what we are evaluating. We are interested in:

1. Whether you can prompt an AI tool to produce correct, relevant results for a specific analytical problem.
2. Whether you understand and can verify the outputs the AI produces (e.g., does the volatility calculation make sense? are the CSV columns correct?).
3. Whether you can diagnose and correct errors in AI-generated code.



4. The overall **quality and correctness** of your final outputs — not just that code runs, but that the results are analytically sound.

You may also consult the internet, documentation, forum posts, and printed material as you would in normal on-the-job work. **Please do not consult other people directly during the exercise.**

Programming Language

You may use either R or Python to complete the exercise. R is the language used most within the cQuant Analytics Team today, but use whichever language lets you produce the best response. Your code will be reviewed for clarity and correctness, so please employ best practices for readability regardless of how it was produced.

Name your main script **main.R** or **main.py** and place any system-specific parameters (file paths, required packages/libraries, etc.) at the top so a cQuant evaluator can update them easily. Please verify your code runs end-to-end before submitting.

Thank you for your interest in cQuant.io and good luck!